Breast Cancer Analysis

Name

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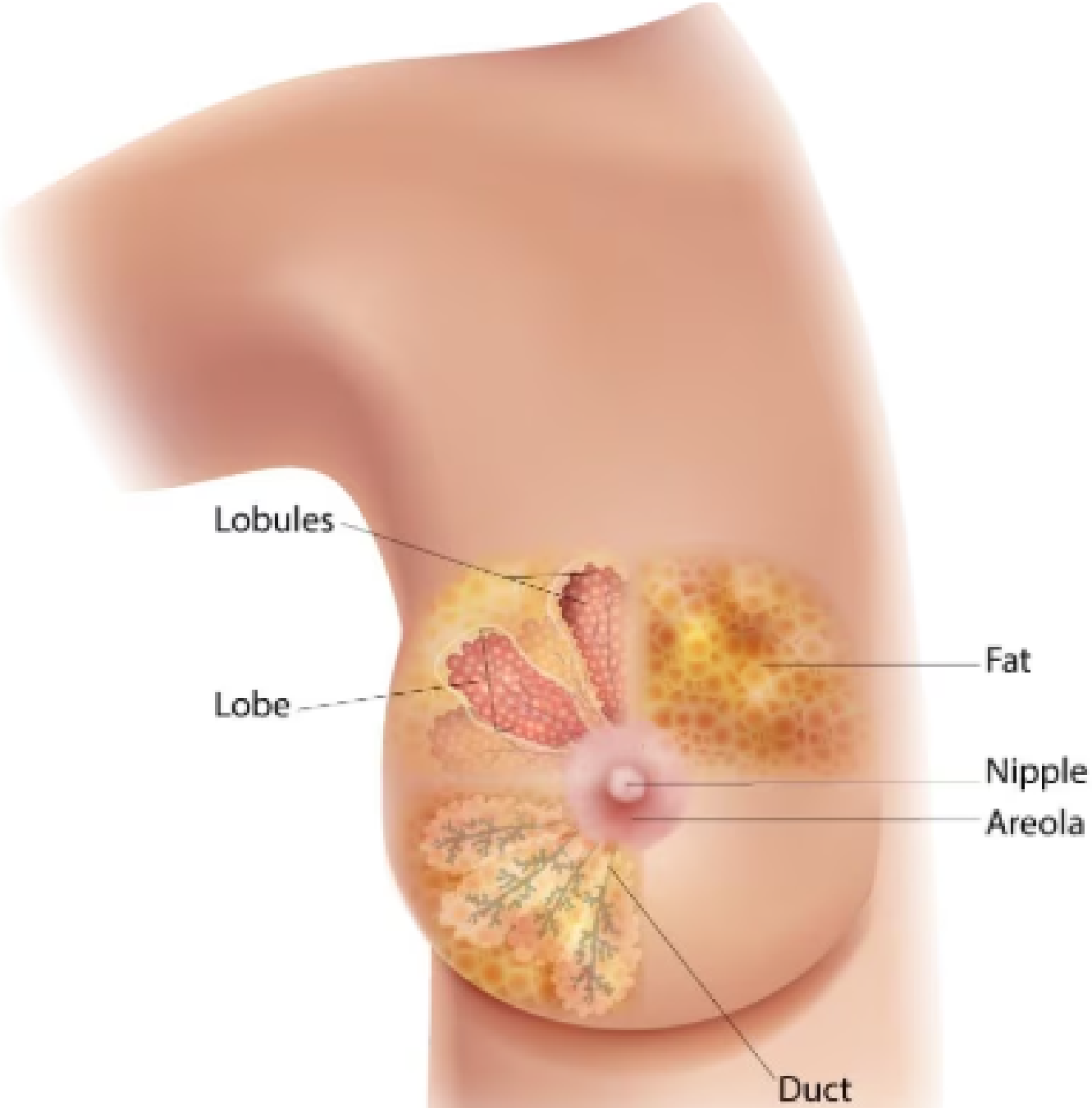
Introduction

- **Significance of Breast Cancer as a Global Health Issue**
  - In 2022, approximately **2.3 million women** were diagnosed with breast cancer worldwide.
  - This represents about **one in four** of all cancers in women.
  - In 2022, **670,000 women** died from breast cancer worldwide.
  - In 2020, breast cancer was the cause of **16%** of cancer deaths in women.
  - Breast cancer incidence has increased by more than **20%** since 2008.
  - Breast cancer incidence is growing in many parts of the world, especially in transitioning countries.
  - In 2019, **25,100 men** were diagnosed with breast cancer globally.

Breast Cancer Overview Image



Brief intro to types of breast cancer

Benign vs. Malignant Tumors

Category	Description
Benign Tumors	Non-cancerous, usually slow-growing, doesn't spread, usually not life-threatening.
Malignant Tumors	Cancerous, can grow rapidly, invade surrounding tissues, spread to other body parts, potentially life-threatening.

Brief intro to types of breast cancer

Invasive Breast Cancers: Types

Feature	**Invasive Ductal Carcinoma (IDC)**	**Invasive Lobular Carcinoma (ILC)**
Frequency	Most common type of breast cancer (70-80%).	Second most common (10-15%).
Origin	Starts in the milk ducts.	Starts in the milk lobules.
Presentation	May form a lump and may spread to other tissues.	May not typically form a lump; presents as a thickening and may affect both breasts.

Project Motivation

- Chose this topic because breast cancer is a major issue.
- Goals of the project:
  - Analyze key datasets related to breast cancer.
  - Apply statistical methods to uncover significant insights.
  - Inform strategies for early detection and treatment.

Dataset Introduction

Source of the Dataset:

This breast cancer domain was obtained from the University Medical Centre, Institute of Oncology, Ljubljana, Yugoslavia. This is one of three domains provided by the Oncology Institute that has repeatedly appeared in the machine learning literature.

Citations:

Zwitter, M. & Soklic, M. (1988). Breast Cancer [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C51P4M

Download Link:

https://archive.ics.uci.edu/dataset/14/breast+cancer

Breast Cancer Diagnosis Data Preview

id	diagnosis	radius_mean	texture_mean
842302	M	17.990000	10.380000
842517	M	20.570000	17.770000
84300903	M	19.690000	21.250000

perimeter_mean	area_mean	smoothness_mean	compactness_mean
122.800000	1001.000000	0.118400	0.277600
132.900000	1326.000000	0.084740	0.078640
130.000000	1203.000000	0.109600	0.159900

Breast Cancer Diagnosis Data Preview (Part 3)

concavity_mean	concave points_mean	symmetry_mean
0.300100	0.147100	0.241900
0.086900	0.070170	0.181200
0.197400	0.127900	0.206900

fractal_dimension_mean	radius_se	texture_se
0.078710	1.095000	0.905300
0.056670	0.543500	0.733900
0.059990	0.745600	0.786900

Breast Cancer Diagnosis Data Preview (Part 4)

perimeter_se	area_se	smoothness_se
8.589000	153.400000	0.006399
3.398000	74.080000	0.005225
4.585000	94.030000	0.006150

compactness_se	concavity_se	concave points_se
0.049040	0.053730	0.015870
0.013080	0.018600	0.013400
0.040060	0.038320	0.020580

Breast Cancer Diagnosis Data Preview (Part 5)

symmetry_se	fractal_dimension_se	radius_worst
0.030030	0.006193	25.380000
0.013890	0.003532	24.990000
0.022500	0.004571	23.570000

texture_worst	perimeter_worst
17.330000	184.600000
23.410000	158.800000
25.530000	152.500000

Global Cancer Data Preview

Label	Cancer code	Country code (ISO/UN)	Alpha-3 code	Sex
Lip oral cavity	1	900	N.A.	0
Salivary glands	2	900	N.A.	0
Oropharynx	3	900	N.A.	0

Type	ASR (World)	Crude rate	Cumulative risk	Total
0	4.000000	4.940000	NaN	389846
0	0.560000	0.700000	NaN	55083
0	1.100000	1.350000	NaN	106400

Breast Cancer Statistics by Country Preview

Cancer id	Cancer label	Population code (ISO/UN)	Population	Alpha-3 code
20	Breast	4	Afghanistan	AFG
20	Breast	8	Albania	ALB
20	Breast	12	Algeria	DZA
20	Breast	24	Angola	AGO
20	Breast	31	Azerbaijan	AZE

Sex	Type	ASR (World) per 100 000
0	0	29.440000
0	0	51.140000
0	0	61.870000
0	0	29.430000
0	0	32.900000

Breast Cancer Continent Mortality Data Preview

Continents	Cancer code	Country code (ISO/UN)	Alpha-3 code	Sex
Africa	20	903	N.A.	0
Latin America and the Caribbean	20	904	N.A.	0
Northern America	20	905	N.A.	0

ASR (World)	Crude rate	Cumulative risk	Total
19.160000	12.970000	NaN	91252
13.210000	17.700000	NaN	59876
12.320000	26.380000	NaN	49744

Statistical Measures and Tools - Introduction

Statistical Tools for Data Analysis

In this section, we will introduce the statistical measures, concepts, and visualization tools employed to analyze the breast cancer datasets. These tools help us to:

- Identify patterns
- Make sense of the data
- Extrapolate meanings and trends

Statistical Measures and Tools (Part 1)

Categories of Statistical Tools Used: Descriptive Statistics (Part 1)

- Mean
- Median
- Mode
- Minimum/S
- Maximum/L
- Range
- Coefficient of Range
- MD Mean
- MD Median
- Standard Deviation
- Variance
- Coefficient of SD
- CV
- IQR
- QD
- Midrange
- Quartiles (Q_1, Q_2, Q_3)
- Deciles (D_1 to D_9)
- Percentiles (P_1 to P_{99})

Statistical Measures and Tools (Part 2)

Categories of Statistical Tools Used: Statistical Formulas

- Central Moments (μ_r)
- Raw Moments (μ'_r)
- Skewness (Moments)
- Kurtosis (Percentiles)
- Excess Kurtosis (Moments)
- Covariance
- Pearson Skewness
- Bowley Skewness
- Correlation
- Regression byx (b_{yx})
- Regression bxy (b_{xy})
- Chebyshev's Inequality
- Normal PDF

Statistical Measures and Tools (Part 3)

Categories of Statistical Tools Used:

Statistical Concepts and Tools

- PMF
- PDF
- CDF
- Expected Value
- Variance
- R-squared

Inferential Statistics

- Regression Analysis

Visualization tools

- Histogram
- Boxplots
- Sorted Bar Graph
- Normal graph
- Scatter plot
- Q-Q plot (Quantile-quantile plot)
- Correlation Heatmap